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Symposium/Special Issue

Journal of Diabetes Science and Technology
2026, Vol. 20(3) 673–683
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DOI: 10.1177/19322968261431860
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Use of Continuous Glucose Monitoring With Machine Learning to Identify Metabolic Subphenotypes and Inform Precision Lifestyle Changes

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Abstract

The classification of diabetes and prediabetes by static glucose thresholds obscures the pathophysiological dysglycemia heterogeneity, primarily driven by insulin resistance (IR), β-cell dysfunction, and incretin deficiency. This review demonstrates that continuous glucose monitoring (CGM) and wearable technologies enable a paradigm shift toward non-invasive, dynamic metabolic phenotyping. We show evidence that machine learning models can leverage high-resolution glucose data from at-home, CGM-enabled oral glucose tolerance tests to accurately predict gold-standard measures of muscle IR and β-cell function. This personalized characterization extends to real-world nutrition, where an individual's unique postprandial glycemic response (PPGR) to standardized meals, such as the relative glucose spike to potatoes versus grapes, could serve as a biomarker for their metabolic subtype. Moreover, integrating wearable data reveals that habitual diet, sleep, and physical activity patterns, particularly their timing, are uniquely associated with specific metabolic dysfunctions, informing precision lifestyle interventions. The efficacy of dietary mitigators in attenuating PPGR is also shown to be phenotype-dependent. Collectively, this evidence demonstrates that CGM can deconstruct the complexity of early dysglycemia into distinct, actionable subphenotypes. This approach moves beyond simple glycemic control, paving the way for targeted nutritional, behavioral, and pharmacological strategies tailored to an individual's core metabolic defects, thereby paving the way for a new era of precision diabetes prevention.

Keywords

diabetes, CGM, wearables, artificial intelligence

Introduction

Significance of Diabetes and Need to Define Heterogeneity

Diabetes affects 38.1 million, or 14.7%, of all US adults aged 18 years or older, and another 38 million, or 38%, have prediabetes, of whom less than 20% are aware.[1] Diabetes and prediabetes are classified based on the degree of glucose elevation, but this does not reflect the multiple physiological pathways that can lead to elevations in blood glucose. Specifically, insulin resistance (IR), beta-cell failure, incretin deficiency, and unrestrained hepatic glucose production (hepatic IR) all contribute. If it were possible to determine which process/processes were the predominant contributors to dysglycemia in a given individual, targeted therapy could be employed, potentially yielding more successful diabetes prevention and treatment than following a one-size-fits-all

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approach. Historically, both lifestyle and medical interventions have been recommended based on the degree of overweight and glucose elevation without regard for (1) interindividual variability in pathophysiology, (2) risk trajectory, or (3) anticipated individualized treatment response. With new technologies, such as continuous glucose monitoring (CGM), it may be possible to characterize and subtype early stages of dysglycemia that reflect/predict these important characteristics, thus significantly impacting human health. In the following review, we outline several novel aspects of CGM and other wearable devices that make inroads into better characterizing metabolic heterogeneity of adults with early dysglycemia, and highlight the interactions between metabolic physiology and response to specific lifestyle interventions.

Limitations of Current Diabetes Classification

Diabetes and prediabetes are diagnosed based on glucose cutoffs.[2-5] Thus, diagnosis is based on an elevated plasma glucose without regard for the mechanism that led to the elevation. We now know that many pathophysiological pathways contribute to glucose elevation—and yet these are lumped together under the umbrella of diabetes. Current diabetes classification of non-pregnant adults is limited to type 1, type 2, and other specific forms.[2-5] Of all prevalent cases of diabetes, type 2 represents >95%. It is becoming clear that not all type 2 diabetes patients follow the same clinical trajectory, with some more prone than others to developing deterioration of glucose control, cardiovascular events, nephropathy, retinopathy, or fatty liver disease.[6-9] Similarly, it is also known that not all patients respond equally well to the same diabetes medications.[10] Thus, there is a movement toward subclassifying type 2 diabetes to better reflect the heterogeneity and inform targeted treatments. Methods for subtyping diabetes/prediabetes range from genetic[11] to clinical/laboratory biomarkers.[12] We recently proposed that CGM represents a novel and accurate method that can be used to subtype the early dysglycemic state according to underlying metabolic physiology.[13] This method has the advantage over genetic subclassification of reflecting the current clinical state which can change as a result of contributory environmental factors (eg, body weight, physical activity), and over both genetic and clinical/demographic factors in degree of accuracy.[13] It does not require laboratory tests or measurement of waist circumference, which pose potential obstacles. Furthermore, because only a single CGM sensor is needed, the cost and burden are minimal, and because it does not require access to a health care system, it bypasses socioeconomic and geographic barriers that might preclude diagnosis/subphenotyping in a standard health care system that is relatively inaccessible to low income and/or geographically remote populations. Finally, this method is amenable to tracking changes or stability in metabolic health over time, as demonstrated in the relative stability of a five-point oral glucose tolerance test (OGTT) glucose trajectory over three years.[14]

Overview of Why Postprandial Glucose Pattern Would Be Good for Metabolic Phenotyping Based on Dynamic Features of the Glucose Curve

Glucose elevations following an oral nutrient challenge have been used in clinical practice since the development of the first OGTT in 1917.[15] While the use of this test has changed over time, early in its evolution, post-glucola metrics, including the one-hour glucose, the integrated area-under-the-curve glucose, whether the shape of the glucose curve from a five-point test (0, 30, 60, 90, and 120 minutes) was mono- or biphasic or unclassified was used to identify both underlying metabolic features of IR and beta-cell dysfunction, as well as individuals at highest risk of progressing to diabetes.[16-30] Later, mathematical approaches, such as functional component analysis,[31] and latent trajectory analysis [14] were explored as biomarkers of physiologic subtypes of diabetes. Due to the burden of the five-point OGTT, however, this was soon abandoned in favor of the two-point (0 and 120 minutes) test with use of values for both fasting and 120 minutes as diagnostic thresholds for diabetes based on the correlation with development of retinopathy,[32-34] and use of more complex algorithms or "shape of the curve" glucose for subphenotyping diabetes or prediabetes was not pursued.

With the advent of CGM, the availability of a more nuanced shape of the curve merits deeper exploration of glucose patterns as a potential subphenotyping strategy that reflects both underlying physiology and risk of progressing to diabetes and related metabolic complications, such as cardiovascular disease.

Predicting Metabolic Subphenotypes With OGTT Glucose Time-Series and Machine Learning Approach

Predicting Metabolic Subphenotypes Using Glucose Time-Series Data and Machine Learning Approach From Frequently Sampled OGTT

Initial efforts to identify metabolic subphenotypes from the glucose shape of the cure utilized a five-point OGTT latent class trajectory analysis in 5861 patients without diabetes from the Danish Inter99 cohort.[6] In this approach, patients underwent metabolic characterization using the euglycemic clamp for insulin sensitivity and the intravenous glucose tolerance test for insulin secretion with a calculated disposition index to measure beta-cell function; glucose shape of the curve was determined from plasma at time 0, 30, 60, 90, and 120 minutes during standardized 75 g, two-hour OGTT. Four different glucose patterns were identified and found to predict both baseline metabolic profile and

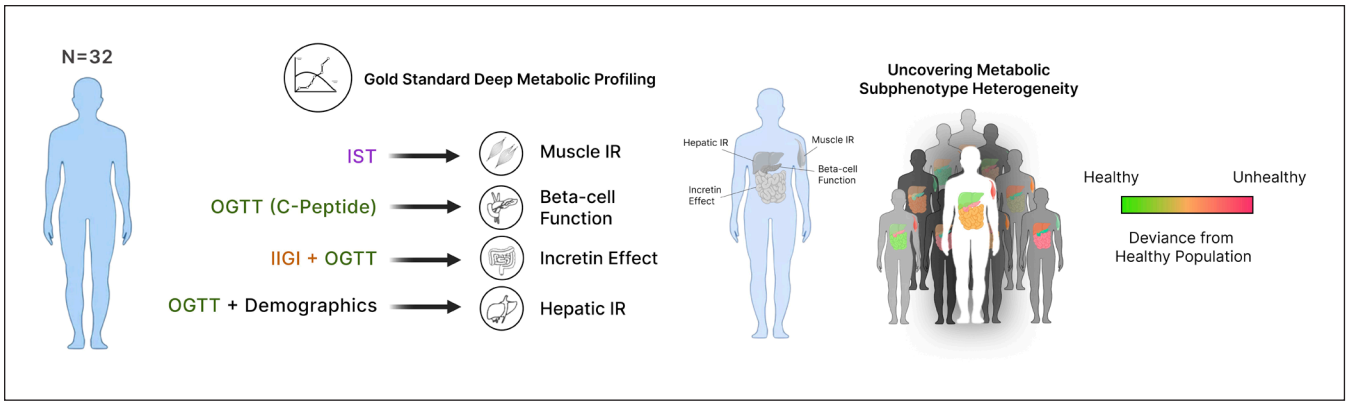


Figure 1. Deep metabolic profiling via gold-standard quantitative tests were used in[13] to evaluate: (1) muscle insulin resistance via the insulin suppression test (IST), (2) beta-cell function by calculating the disposition index from an oral glucose tolerance test (OGTT) using C-peptide deconvolution, (3) incretin effect using an isoglycemic intravenous glucose infusion (IIGI) and an OGTT, and (4) hepatic insulin resistance using a validated formula based on OGTT insulin levels and demographic information.

subsequent clinical risk for diabetes, retinopathy, nephropathy, cardiovascular disease, and all-cause mortality over a 13-year follow-up period.[6] In a separate study, these curve profiles were also evaluated for stability over time: individuals subgroup based on glucose curve remained the same in 55% over three years; moved to a more favorable profile in 19%, and to a less favorable profile in 27%, which was associated with increase in body mass index (BMI) and related metabolic risk markers.[14]

Building on this body of data, a group of researchers at Stanford University introduced a more complex framework for predicting metabolic subphenotypes using glucose time-series data from a 16-point, three-hour, 75 g OGTT.[13] In this study, metabolic physiologic profile was deeply phenotyped using gold-standard metabolic tests in a cohort of individuals with normoglycemia or prediabetes (Figure 1). These rigorous tests precisely quantified four key physiological processes: muscle IR (measured by modified insulin suppression test [steady-state plasma glucose, SSPG], hepatic IR (inferred by a validated index), β-cell dysfunction (measured via disposition index), impaired incretin function quantified by calculating relative insulin secretion during OGTT vs isoglycemic intravenous glucose infusion (IIGI)[35]. The study revealed that individuals exhibited diverse combinations of metabolic defects. Specifically, muscle and hepatic IR were highly correlated, accounting for a single or co-dominant metabolic phenotype in 35% of individuals, while β-cell dysfunction and/or incretin deficiency accounted for 42%. The remaining participants did not exhibit these specific dominant drivers. Importantly, these underlying metabolic dysfunctions did not correlate with traditional glycemic measures like glycated hemoglobin (HbA1c), highlighting the inadequacy of current diagnostic approaches to subclassify early stages of dysglycemia.

To translate these complex physiological insights into a scalable diagnostic tool, the team developed a machine learning (ML) framework that utilizes features extracted from a frequently sampled, 16-point plasma glucose curve obtained during OGTTs in a research unit. This dense sampling effectively mimicked the continuous data stream provided by a CGM, offering a "CGM-like" assessment in the research setting. Two main feature extraction approaches were employed: "OGTT_G_Features," encompassing 14 engineered features (eg, area under the curve [AUC], peak glucose, and slopes), and "OGTT_G_ReducedRep," a reduced representation using principal component analysis. This approach demonstrated the superior predictive power of ML models trained on high-resolution plasma glucose data for predicting metabolic subphenotypes compared with current clinical biomarkers, laboratory tests, and even polygenic risk scores.[13,36] Specifically, muscle IR was predicted with a robust area under the curve (auROC) of 0.95 using OGTT_G_ReducedRep, as compared with homeostatic model assessment of insulin resistance (HOMA-IR) with area under the receiver operating characteristic curve (auROC) of 0.77, while β-cell dysfunction and incretin deficiency achieved auROCs of 0.89 and 0.88, respectively, which compared with auROCs of 0.48 and 0.68 for HOMA-ß and one-hour glucagon-like peptide and glucose-dependent insulinotropic polypeptide, respectively (Figure 2). While prediction for hepatic IR was promising (auROC = 0.84), other surrogate markers, such as HOMA-IR[37] and Matsuda index,[38] showed comparable or superior performance.

Predicting Metabolic Subphenotypes From CGM During At-Home OGTT

To assess the feasibility of using at-home CGM testing to predict muscle IR and β-cell function, Metwally et al recruited 29 participants who underwent both gold-standard clinical tests (the SSPG test for IR and a 16-point OGTT with C-peptide deconvolution for β-cell function) and two standardized OGTTs at home while their glucose levels were recorded by a CGM (Dexcom G6 pro) (Figure 3). To ensure

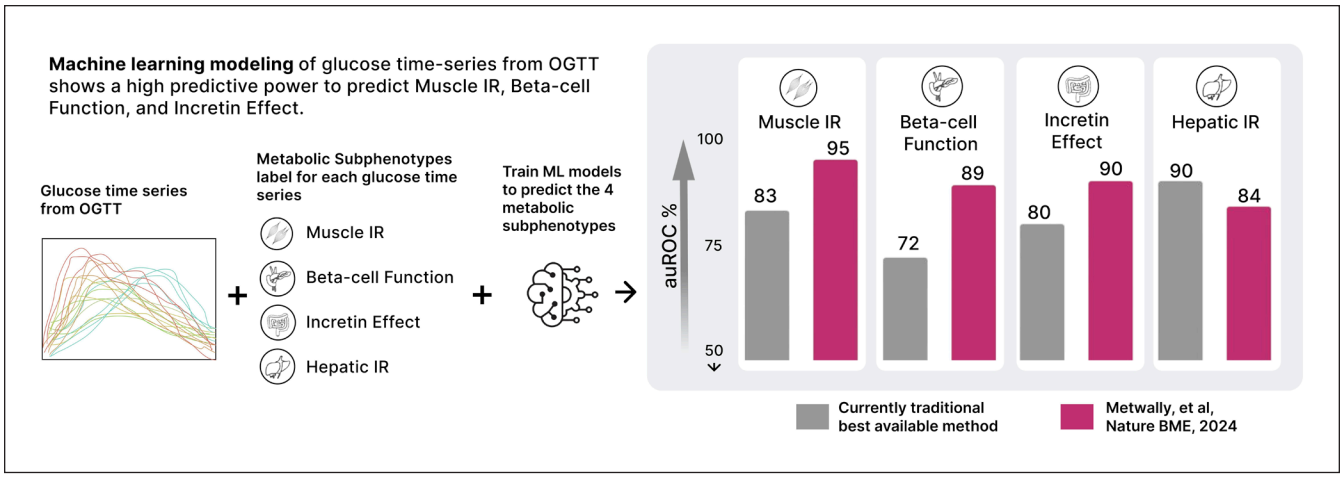


Figure 2. Robust performance in predicting muscle IR, beta-cell dysfunction, and incretin effect using glucose time-series compared with currently used surrogate markers.

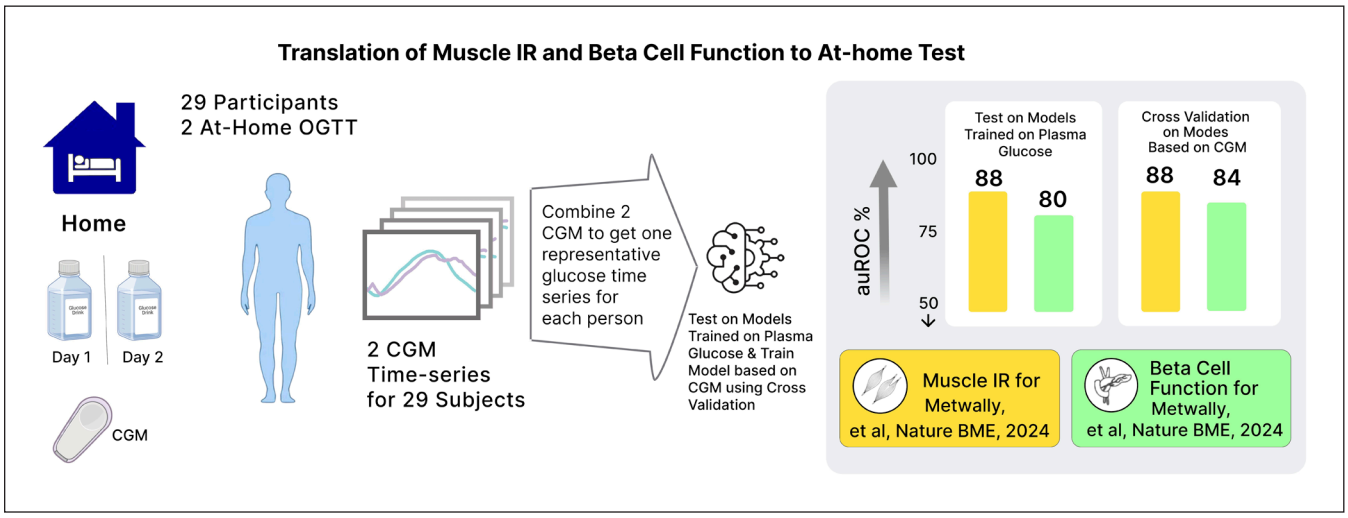


Figure 3. Study design of the at-home OGTT test via CGM to predict muscle IR and β-cell function. Participants underwent gold-standard testing at the research unit for insulin resistance (insulin suppression test) and B-cell function (16-point OGTT with C-peptide deconvolution adjusted for SSPG and expressed as DI) as described, as well as two OGTTs administered at home under standardized conditions during which glucose patterns were captured by a CGM within a single 10-day session (Dexcom G6 pro).

reproducibility, scalability, and minimize environmental influences, participants were instructed to perform the two at-home OGTTs within a single 10-day CGM sensor session after an overnight fast. They consumed a 75-g glucola drink identical to the in-clinic procedure, remaining seated for three hours with no intake other than water. Detailed instructions were provided regarding diet and physical activity before the test, avoiding celebratory or restrictive eating, strenuous exercise, and late-night meals. This meticulous standardization aimed to control variability, a known challenge with OGTTs. The study found high concordance between glucose time-series data obtained from (1) in-clinic plasma sampling and simultaneous in-clinic CGM measurements (r = 0.81), (2) two repeated at-home OGTTs using CGM (r = .86), indicating strong reproducibility of the at-home method, and (3) in-clinic venous measurements and the mean of two at-home CGM OGTTs (r = 0.80). These high correlations indicate that CGM measurements performed at home are similar to those obtained in a clinical research unit and are highly reproducible when conducted under standardized conditions.

The ML framework developed in the first part of the study (using frequently sampled glucose time-series from OGTT done at the research unit), when applied to CGM-generated glucose curves from at-home OGTTs, accurately predicted metabolic subphenotypes. Insulin resistance was predicted with an AUC of 88%. Similar to the plasma-based OGTT at research unit, at-home CGM curves from at-home OGTT clearly showed differences between insulin-resistant and insulin-sensitive (IS) individuals, with IR individuals

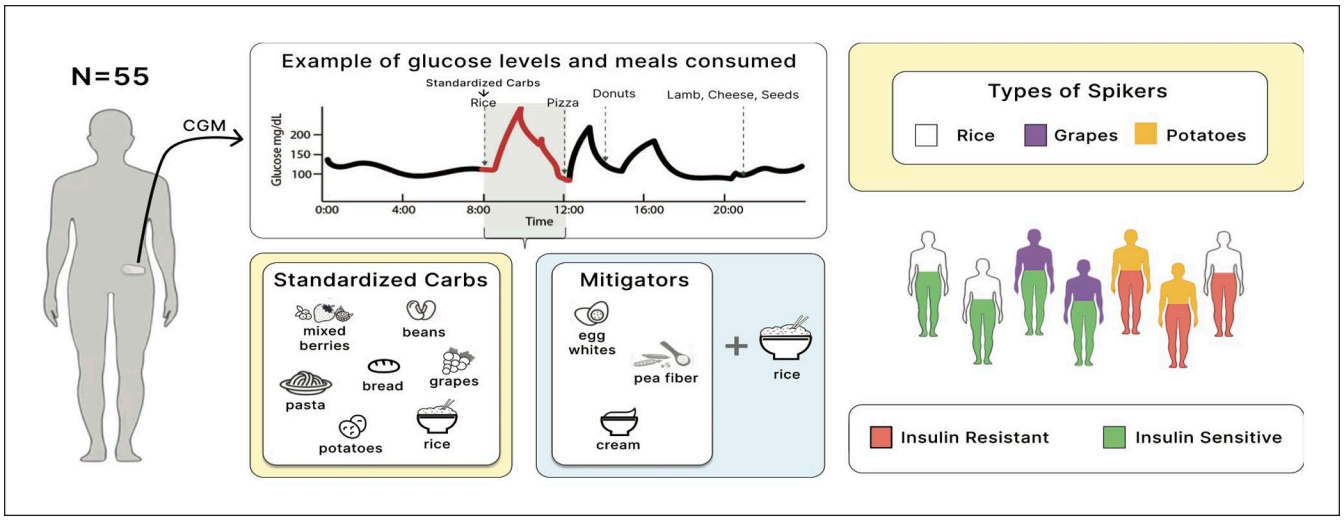


Figure 4. Study design of the standardized meal study. Fifty-five participants consumed seven different standardized carbohydrate meals and three mitigator foods with rice with replicates while wearing CGM. Participants were grouped by their highest response to carbohydrate and measured metabolic subphenotypes.

exhibiting higher glucose peaks, broader curves, and higher AUCs. Moreover, β-cell dysfunction was predicted with an AUC of 80% using the mean of two home CGM curves. This performance further improved to an AUC of 84% when using cross-validation based solely on CGM data.

In summary, this novel approach demonstrates the feasibility of conducting an at-home OGTT with a CGM, which provides a practical, accurate, and scalable method for identifying distinct metabolic subphenotypes of T2D, such as IR and beta-cell function. It removes the patient burden associated with multiple blood draws over a three-hour window in a clinical setting, making the OGTT more feasible and accessible for widespread use. A notable limitation of the current metabolic subphenotyping framework is that the influence of concurrent medications on dynamic glucose patterns remains unknown. Future research is required to evaluate the robustness of subphenotyping models across diverse medication populations to ensure their broad clinical utility.

Predicting Metabolic Subphenotypes Using CGM at Home With Real Food Challenges

Variation in individuals' postprandial glycemic response (PPGR) to the same meals has been clearly demonstrated[39,40]. The determinants of this variability are incompletely characterized, but several studies have implicated microbiome, genetics, meal composition and timing, and other patient characteristics as contributors, and models to predict PPGR based on these inputs have shown reasonable success (r = 0.68-0.77).[39,40,41,42] A subsequent study from our group demonstrated that interindividual variability on PPGR was a function of underlying metabolic phenotype.[42] In this study, 55 participants without diabetes were recruited for deep metabolic subphenotyping as measured by gold-standard tests in the metabolic research unit (average age of 55.2 years, BMI of 25.6 kg/m^2, HbA1c of 5.6%, and relatively equal sex composition "27 females, 24 male, 4 unknown"). Participants consumed seven different carbohydrate meals in replicate with the same amount of total carbohydrates (50 g) on different days in the morning after an overnight fast and under standardized conditions to remove environmental confounders (eg, coffee, exercise, late-night snack, eating out the night prior) while wearing a Dexcom G6 CGM, which were placed on the abdomen, and operated in blinded mode. The study was approved by the Stanford Institutional Review Board (IRB #43883). The seven carbohydrate meals included rice, bread, potatoes, pasta, black beans, mixed berries (blackberries, strawberries, and blueberries), and grapes (Figure 4).

The researchers found that different participants had different responses to the same carbohydrate meals and had different carbohydrates producing the highest PPGRs as measured by AUC (>baseline) and delta glucose.[42] Participants were stratified into "spiker" type based on which carbohydrate meal produced the highest average PPGR. For instance, when grouped according to delta glucose peak, rice-spikers were the largest group (35%), followed by bread-spikers (24%) and grape-spikers (22%) (Figure 5)

Interestingly, PPGR to the same foods differed significantly according to metabolic subphenotype. The PPGR to potatoes and pasta, but not other carbohydrates, differed significantly as a function of IR, and the PPGR to potatoes differed significantly in those with normal or insufficient beta-cell function (Figure 6A). The relative elevation in PPGR to potato in comparison to grapes (potato-grape ratio) differed significantly between the metabolically healthy (IS) and unhealthy (insulin-resistant), with almost no overlap, suggesting it might be used as a biomarker for metabolic subphenotype (Figure 6B). While more work will be needed

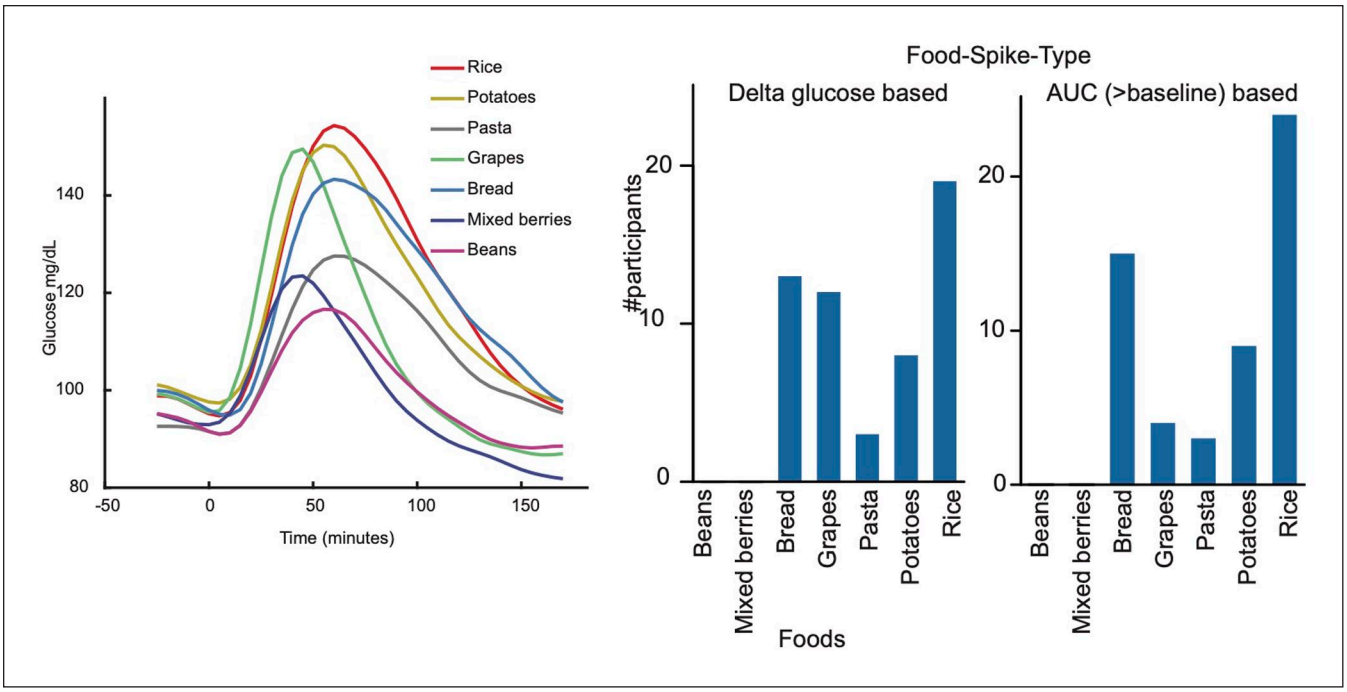


Figure 5. Postprandial glycemic responses to different carbohydrates. Left: mean CGM curves after different meals. The *X*-axis is time, with the food log consumption time as 0, and the *Y*-axis is glucose level. Right: number of participants classified to each spiker type as defined by both delta glucose peak and AUC (>baseline). The *X*-axis indicates different carbohydrates, and the *Y*-axis is the number of participants for whom a given carbohydrate produced the highest spike.

to validate this biomarker in larger populations, it is the first time a glycemic response to real food has been demonstrated to predict underlying metabolic physiology. Of note, not only were "potato spikers" more likely to be insulin-resistant and/or beta-cell deficient, they also exhibited higher HbA1 and fasting blood glucose, indicating a worse metabolic state, in particular, compared with grape-spikers. Figure 6C and 6D illustrates how the relative elevations in delta glucose peak and AUC, derived from all tested standardized meals, relate to IR and beta-cell function. In addition, Asian participants were more likely than non-Asian participants to be rice-spikers. Bread-spikers exhibited significantly higher systolic and diastolic blood pressure than other carb-response types.

The results of this study demonstrate for the first time an association between metabolic subphenotype and an individual's relative PPGR to some carbohydrates over others. This may have implications for both personalized clinical nutrition and diabetes prevention (avoid potatoes if you are insulin-resistant or beta-cell deficient).

Associations Between Metabolic Subphenotypes and Real-World Lifestyle Patterns

Diet, sleep, and physical activity are core lifestyle behaviors that can be modified to support metabolic health.[43-47] Their effects on glucose regulation have been widely studied, primarily in relation to glucose levels.[43] Specifically, certain dietary components (eg, nutrients and food groups) and broader dietary patterns are known to influence fasting blood glucose, hemoglobin HbA1c, and IR.[40,48,49] Similarly, both sleep duration and quality, along with physical activity levels, have demonstrated associations with these glycemic markers.[2-5] However, important gaps remain in our understanding of how these behaviors contribute to the development of metabolic diseases. Notably, few studies have assessed all three lifestyle domains in an integrated manner.

Park et al[50] extended this research by examining beyond the conventional associations between lifestyle behaviors and glycemic markers to investigate how these behaviors relate to distinct metabolic characteristics (herein, "metabolic subphenotypes"). They hypothesized that lifestyle factors would be associated not only with standard glycemic outcomes, such as fasting glucose, A1C, and OGTT results, but also with specific physiological dysfunctions, including IR, beta-cell dysfunction, and compromised incretin response. A mobile food logging application was used for dietary data collection, and sleep and physical activity data were collected through wearable devices. The extensive multimodal data from three lifestyle domains, along with demographic covariates, were integrated into predictive models for distinct metabolic subphenotypes. Notably, they identified unique sets of lifestyle predictors in each model (Figure 7). Age and BMI were shared predictors across several dysfunctional metabolic subphenotypes, such as incretin dysfunction, beta-cell

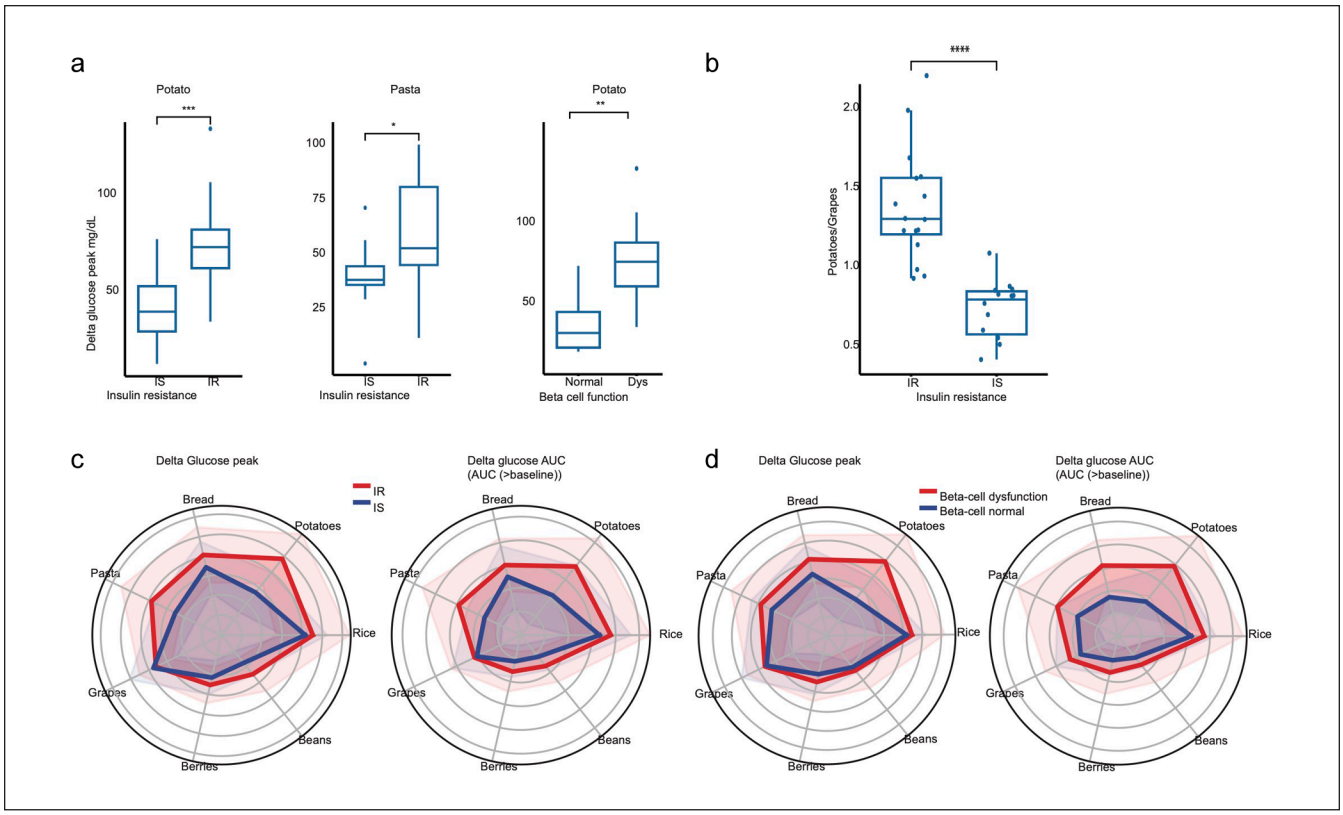


Figure 6. Differential responses to potato and pasta across people with different metabolic subphenotypes. (a) Delta glucose peaks after eating potatoes and pasta are compared between groups with IR and IS, and delta glucose peaks after eating potatoes are compared between groups with normal and dysfunctional (dys) beta-cell functions. (b) The ratio between the response (delta glucose) to potato and that to grapes are compared between groups with IR and IS. (c) Average delta glucose peak and AUC (>baseline) between participants with IR and IS. (d) Average delta glucose and AUC (>baseline) between participants with different beta-cell functions (normal and dysfunctional). The solid line is the average value, and the shade is one *SD*.

dysfunction, and IR in all specific tissues (muscle, adipose tissue, and liver). Among lifestyle factors, total exercise duration predicted normal beta-cell function as well as insulin sensitivity in muscle and adipose tissues. However, it was not a significant predictor of hepatic insulin sensitivity. Distinct sleep characteristics were also differentially associated with subphenotypes: wake-up time was associated with muscle IR and incretin function, whereas sleep latency (ie, the time to transition from full wakefulness to sleep onset) was related to IR in adipose tissue. In the dietary domain, fiber intake showed protective associations with both adipose and hepatic IR, but no dietary factor significantly predicted muscle IR. Interestingly, hepatic insulin sensitivity appeared to be mostly associated with dietary factors, including not only dietary composition but also meal timing.

These findings highlight the complexity of metabolic regulation and its variation across tissues. This work also emphasizes the potential of prevention strategies that are tailored to an individual's habitual lifestyles and metabolic susceptibility. A precision-based approach may offer more effective results in preventing the onset of type 2 diabetes.

Leveraging CGM to Evaluate Dietary Mitigators of PPGR

High PPGR is a risk factor for CVD and all-cause mortality independent of fasting blood glucose and HbA1c [51,52]. Therefore, reducing PPGR to meals (mitigation) is one of the goals of nutrient intervention. In the work of Wu et al,[42] in addition to the seven standardized carbohydrate meals, three mitigators containing fiber (10 g, via pea fiber), protein (10 g, via egg white), and fat (15 g, via cream) were also tested for their ability to attenuate the PPGRs to rice, which on average produced the greatest glucose elevation in their study. To determine the effects of the suppression mitigator, we calculated the difference between the peak glucose change (delta glucose) of the rice-plus-mitigator group and the rice-only group, then normalized the result against the rice-only values. Thirty-two participants consumed fiber, protein, or fat mitigators 10 minutes before consuming rice (50 g total carbohydrates). The mitigation effect was calculated as a normalized difference between PPGRs of rice + mitigator and rice. A negative value indicated that consuming the mitigator in front reduced the PPGR compared with rice alone.

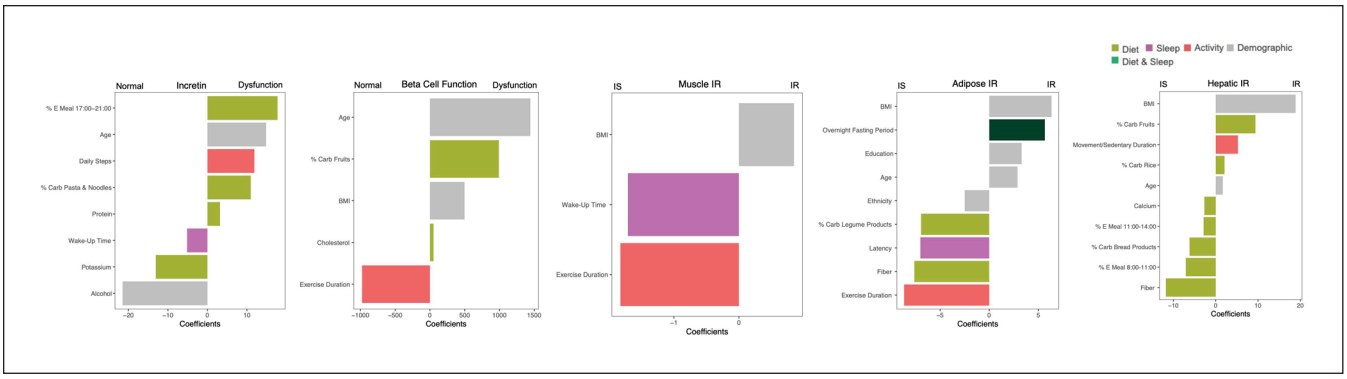


Figure 7. Integrated lifestyle prediction models predicting metabolic subphenotypes. LASSO classification models were built using all lifestyle features, and the coefficients of the final predictors are shown. Classifications include incretin and beta-cell normal function vs dysfunction as well as muscle, adipose, and hepatic insulin sensitivity (IS) vs insulin resistance (IR). Colors denote feature categories. Sex (1 = male, 0 = female) and ethnicity (1 = Caucasian, 0 = non-Caucasian) are numeric variables. Latency is the time from wakefulness to sleep onset. %E Meal and %Carb indicate proportional energy or carbohydrate contribution. Movement/sedentary ratio and education (years) are continuous variables. Nutrients (eg, fiber, sodium) represent daily intakes.

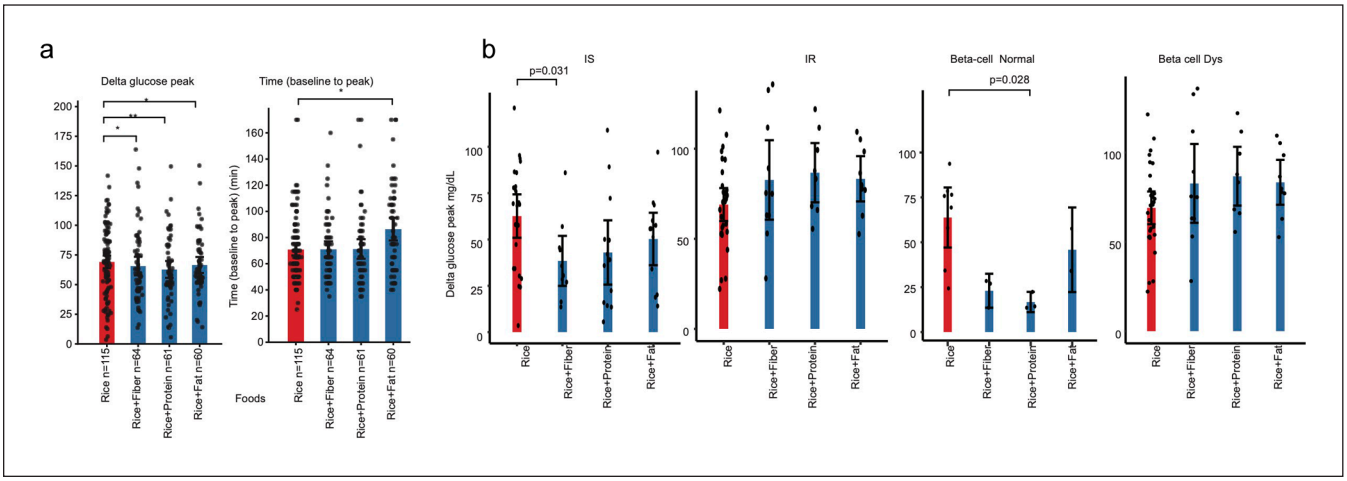


Figure 8. Mitigator effects of fiber, protein and fat in different groups. (a) The effect of mitigators on delta glucose and time from baseline to peak in the whole cohort. (b) The effect of mitigators on delta glucose on the group with IR, IS, normal beta cell, and dysfunctional beta cell. The red columns represent rice meals without mitigators, while the blue columns represent rice meals with mitigators.

Across the entire cohort, all three mitigators produced a modest but statistically significant reduction in PPGR to rice. Fat, but not protein or fiber, delayed the time from the baseline to the peak glucose (Figure 8). Interindividual variability was also observed in mitigation effects. In addition, similar associations between PPGR and metabolic subphenotypes were also observed in the mitigation effect. Individuals with IR presented minimal PPGR reduction with mitigators, whereas individuals with IS exhibited stronger responses to mitigators, with fiber reducing delta glucose significantly. Individuals with normal beta-cell function also had a protein significantly reducing the delta glucose peak.

The results in this study suggest not only the targeted dietary intervention for individuals with different metabolic subtypes but also the need of systematic deep profiling of the digestion and absorption process. In particular, individuals who are insulin-resistant or have beta-cell dysfunction might require additional personalized mitigation solutions rather than the conventional dietary suggestion.

Discussion

The data presented above highlight the potential of CGM to deconstruct the metabolic complexity of early dysglycemia. The multiple datapoints generated by CGM during dynamic conditions, such as oral glucose loads or meal challenges, are amenable to ML approaches that generate predictive algorithms capable of identifying distinct metabolic subphenotypes. Phenotypes, including IR, beta-cell dysfunction, and incretin deficiency have implications for clinical risk,

precision nutrition, and lifestyle interventions that may decrease risk for progression to diabetes. The ability to identify these metabolic phenotypes at home using CGM has benefits with regard to low patient burden (compared with OGTT in a clinical setting) and high patient appeal. Continuous glucose monitoring glucose patterns may also be able to inform precision medical therapy that is targeted toward the predicted physiologic deficit. Future studies should address the stability of glycemic patterns over time and whether the expected shift from an IR pattern to an IS pattern on CGM ensues following interventions that improve insulin sensitivity (eg, weight loss), so that the CGM could be used longitudinally to evaluate changes in metabolic risk profile in the same individual. In the near future, smartwatch data could be integrated with CGM for metabolic subphenotyping prediction[53,54], and integrated into conversational artificial intelligence systems to help users interpret results.[53,55-58] Thus, with the potential of CGM to improve human health still in its infancy, the future for diabetes and metabolic diseases may be brighter as a result of this relatively new wearable technology.

Abbreviations

AUC, area under the curve; auROC, area under the receiver operating characteristic curve; BMI, body mass index; CGM, continuous glucose monitoring; CVD, cardiovascular disease; DI, disposition index; HbA1c, hemoglobin A1c; IIGI, isoglycemic intravenous glucose infusion; IR, insulin resistance; IRB, institutional review board; IS, insulin sensitivity; IST, insulin suppression test; ML, machine learning; OGTT, oral glucose tolerance test; PPGR, postprandial glycemic response; SSPG, steady-state plasma glucose (often measured via the insulin suppression test); T2D, type 2 diabetes.

Acknowledgments

The authors thank Lettie McGuire for her tremendous work on the illustrations used in this work.

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Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Declaration of Conflicting Interests

The authors declared the following potential conflicts of interest with respect to the research, authorship, and/or publication of this article: AAM is an employee of Alphabet and may own stock as part of the standard compensation Package. TM is a member of the scientific advisory board of January AI. MPS is a cofounder and scientific advisor of Xthera, Exposomics, Filtricine, Fodsel, iollo, InVu Health, January AI, Marble Therapeutics, Mirvie, Next Thought AI, Orange Street Ventures, Personalis, Protos Biologics, Qbio, RTHM, and SensOmics. MPS is a scientific advisor of Abbratech, Applied Cognition, Enovone, Jupiter Therapeutics, M3 Helium, Mitrix, Neuvivo, Onza, Sigil Biosciences, Captify Inc, WndrHLTH, Yuvan Research, and Ovul.

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